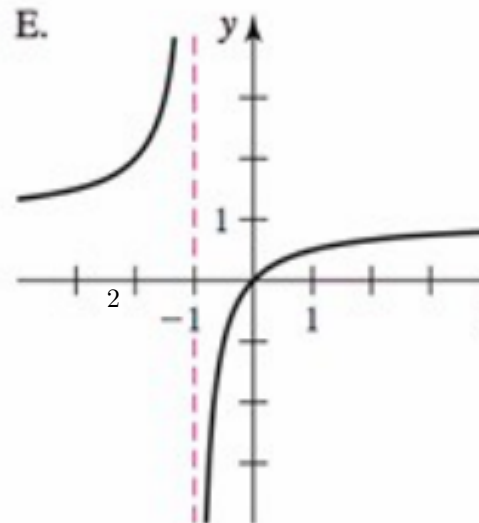
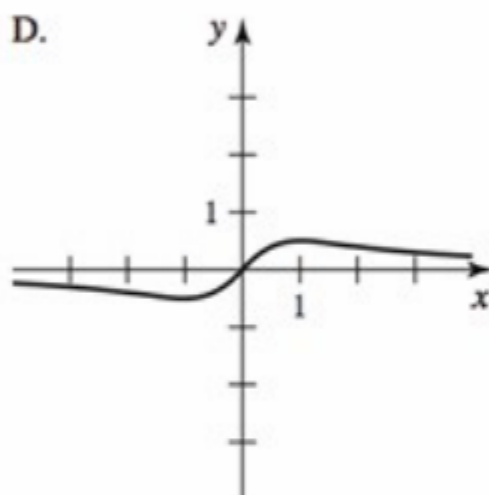
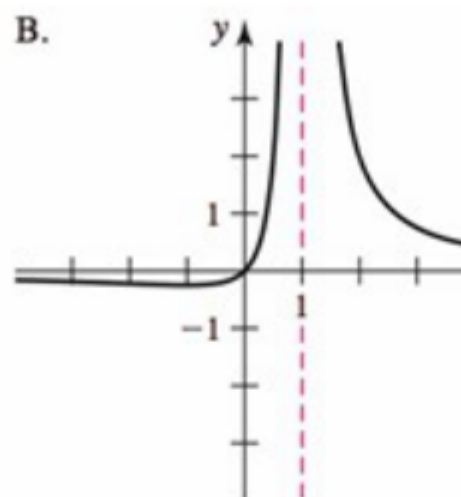
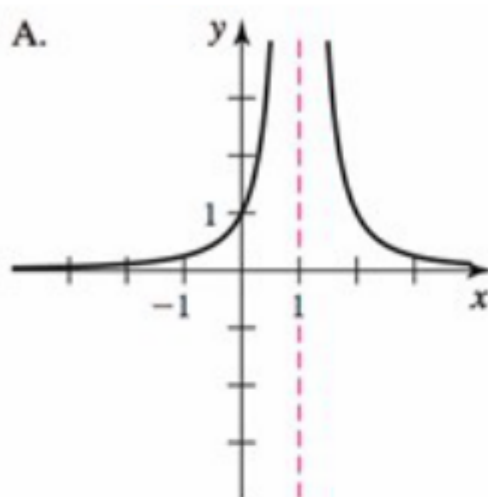


Problem 1

Problem 1

Problem 1 Without using a graphing utility, match each graph of functions in A-F with the algebraic representation of functions in a-f:

Problem 1



- (a) The function f defined by $f(x) = \frac{x}{x^2 + 1}$.

Explanation. Since there is no real number x such that $x^2 + 1 = 0$, we have no candidates for vertical asymptotes. Hence the graph of f should not contain any vertical asymptotes. The only listed graph with no vertical asymptote is graph D.

- (b) The function g defined by $g(x) = \frac{x}{x^2 - 1}$.

Explanation. *Candidates* for vertical asymptotes:

$$x^2 - 1 = 0 \implies x = 1 \text{ or } x = -1.$$

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{x}{x^2 - 1} = \lim_{x \rightarrow 1^+} \frac{x}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator also positive.

Therefore, vertical asymptote at $x = 1$.

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^-} \frac{x}{x^2 - 1} = \lim_{x \rightarrow -1^-} \frac{x}{(x - 1)(x + 1)} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator negative, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $g(0) = 0$, the only listed graph that can match is graph C.

- (c) The function h defined by $h(x) = \frac{1}{x^2 - 1}$.

Explanation. *Candidates* for vertical asymptotes: $x = 1$ and $x = -1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{1}{x^2 - 1} = \lim_{x \rightarrow 1^+} \frac{1}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Problem 1

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^-} \frac{1}{x^2 - 1} = \lim_{x \rightarrow -1^-} \frac{1}{(x - 1)(x + 1)} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $h(0) = -1$, the only listed graph that can match is graph F.

(d) The function a defined by $a(x) = \frac{x}{(x - 1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{x}{(x - 1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Since $a(0) = 0$ the graph is graph B.

(e) The function s defined by $s(x) = \frac{1}{(x - 1)^2}$.

Explanation. Candidate for the vertical asymptote: $x = 1$.

Test of candidate $x = 1$:

$$\lim_{x \rightarrow 1^+} \frac{1}{(x - 1)^2} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, the denominator positive.

Therefore, vertical asymptote at $x = 1$.

Since $s(0) = 1$ the graph is graph A.

(f) The function r defined by $r(x) = \frac{x}{x + 1}$.

Problem 1

Explanation. Candidate for the vertical asymptote: $x = -1$.

Test of candidate $x = -1$:

$$\lim_{x \rightarrow -1^+} \frac{x}{x+1} = -\infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator negative, the denominator positive.

Therefore, vertical asymptote at $x = -1$.

Since $r(0) = 0$ the graph is graph E.